AutoTerm-SST: Zero-Setup Adaptive Terminology Memory for Streaming Speech Translation

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Abstract

We present AutoTerm-SST, an interactive system for terminology-aware streaming speech translation. Live domain talks are dense in rare entities and technical terms, yet requiring users to prepare a glossary before a session creates substantial friction. AutoTerm-SST removes this setup step with an adaptive terminology memory: each streaming chunk retrieves at most 10 glossary candidates from a compact active inventory, and a training-free router decides which domain-specific terminology slice is active. The router combines generated target-translation topic evidence, speech-window domain probes, speech-centroid evidence, and retrieved-term metadata, without running a separate ASR or source-transcript classifier in the production path. The demo exposes retrieved terms, active slices, router decisions, and retrieval latency in a live evidence panel. In a terminology smoke evaluation, the zero-setup working glossary recovers eight of nine gold terms while injecting fewer, less noisy references than a broad 100k open memory; scale probes show warm MaxSim retrieval stays near 60–105 ms (p50–p95) up to million-term memories, and a mixed-domain routing probe follows ACL/medicine domain transitions with no wrong switches. AutoTerm-SST is released as a pluggable RASST-Demo framework with mock and live GPU modes.

1 Introduction

Simultaneous speech translation (SST) is increasingly used for technical talks, lectures, meetings, and live media. Prior SST work studies the quality-latency trade-off of prefix translation (Ma et al., 2019) and standardized streaming evaluation (Ma et al., 2020). Recent omni speech language models improve fluency and long-form context handling (Qwen Team, 2025), but live systems still struggle with term-dense content: named methods, datasets, clinical terms, legal phrases, and financial entities often need specialized translations that are not obvious from local acoustic context alone.

Glossaries are a practical way to control terminology, but they are awkward for live use. A user may not know the exact topic before a talk starts, and asking every speaker or viewer to upload a domain glossary creates friction. A naive alternative retrieves from a broad open memory of hundreds of thousands of terms. AutoTerm-SST retrieves at most 10 glossary candidates per chunk and then filters low-confidence matches; the systems problem is therefore choosing the active inventory and ranking useful terms into a small retrieval cap.

AutoTerm-SST treats this as an active-inventory and ranking problem. The automatic mode starts from a domain-specific initial slice, and a training-free router switches among domain slices when streaming evidence is strong, using broad open memory only as a low-confidence fallback pool.

The result is an interactive streaming SST workbench rather than a new translation model. The framework handles REST/WebSocket transport, session lifecycle, and agent routing; agents own model inference, prompting, batching, and retrieval. The current live agent uses Qwen3-Omni served through vLLM-style continuous batching (Kwon et al., 2023), and the web and Electron interfaces expose retrieved terms, active glossary status, router decisions, and latency alongside translations.

Our contributions are:

- We build a pluggable streaming SST demo framework with a live evidence panel for retrieved terminology, active glossary state, router decisions, and latency.

- We introduce a zero-setup adaptive terminology strategy that routes among domain-specific active slices and caps prompt candidates at a score-filtered top-10.

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- We provide system evidence covering terminology recall, retrieval latency and scale up to million-term memories, mixed-domain routing, and concurrent streaming.

2 Related Work

Streaming speech translation. SST systems must translate partial speech while balancing quality and latency. Prefix-to-prefix policies such as wait-k make this trade-off explicit (Ma et al., 2019), attention-based policies refine read/write decisions (Papi et al., 2023, 2024), SimulEval standardizes evaluation (Ma et al., 2020), and large-scale streaming models target direct simultaneous translation (Seamless Communication, 2023; Zhang et al., 2024). Speech language models make this setting more flexible (Qwen Team, 2025; Ouyang et al., 2025), but live deployment still requires session management, incremental transport, and latency-aware prompting. Unlike existing streaming ST demonstrators, which fix vocabulary context at session start, AutoTerm-SST manages terminology *during* the stream, as a system demonstration rather than a new architecture.

Terminology-aware translation. Terminology control has long been studied in machine translation, usually through user-provided dictionaries, constrained decoding, or training-time exposure to terminology constraints (Hokamp and Liu, 2017; Dinu et al., 2019); terms and named entities are also a known weak point of speech translation specifically (Gaido et al., 2021). Retrieval-augmented generation offers another way to expose non-parametric memory at generation time (Lewis et al., 2020; Khandelwal et al., 2021). These methods assume the domain is known ahead of time. RASST couples a speech-side MaxSim retriever with prompt-injected terminology (Luo et al., 2026); AutoTerm-SST builds on it with a zero-setup interaction pattern, maintaining the active terminology context from retrieval evidence observed during streaming.

LLM serving and demo systems. Modern LLM serving systems provide continuous batching and efficient generation APIs (Kwon et al., 2023; Zheng et al., 2024), but a streaming speech demo also needs microphone/file input, WebSocket transport, session cleanup, live evidence rendering, and mock-mode testing. Translation Canvas (Dandekar et al., 2024) targets offline translation analysis; AutoTerm-SST contributes the live serving layer, keeping model-specific code inside swappable agents.

3 System Overview

AutoTerm-SST is organized as a thin framework plus swappable translation agents (Figure 1). The framework is deliberately small: it serves the web UI, accepts REST control requests, streams audio over WebSockets, tracks session lifecycle, and routes each session to an agent selected by `agent_type`. The framework does not know about model prompts, retrieval, batching, KV cache, or glossary construction.

Agent contract. Agents implement a small contract: `open_session`, `submit_audio`, `on_control`, `close_session`, `describe`, and `health`. The same UI and wire protocol serve a deterministic mock agent for development, a legacy InfiniSST scheduler agent (Ouyang et al., 2025), or the RASST/Qwen3-Omni agent (Luo et al., 2026) used in the live GPU demo. The boundary is backend-agnostic: the live demo uses in-process vLLM (Kwon et al., 2023), and the same protocol can serve external SGLang-compatible backends (Zheng et al., 2024) without changing the terminology memory, router, or UI.

Streaming agent. For each audio chunk, the OmniAgent can run MaxSim terminology retrieval over the active slices, run the hybrid active-slice router, build a `term_map`, and call the in-process vLLM backend. In `auto_working`, the retriever returns up to 10 ranked glossary candidates and filters low-confidence matches before the PromptBuilder constructs the prompt. The agent emits partial translations plus structured metadata, including retrieved terms, retrieval latency, active slices, prompt-reference counts, and router decisions.

Evidence protocol. The default WebSocket output remains plain text for backward compatibility. A JSON mode returns `{type, text, meta}`, where `meta.references` drives the live evidence panel and `meta.topic` / `meta.topic_router` describe the adaptive glossary state. This evidence protocol is central: users see which terms were retrieved and whether the router stayed, switched, or deferred.

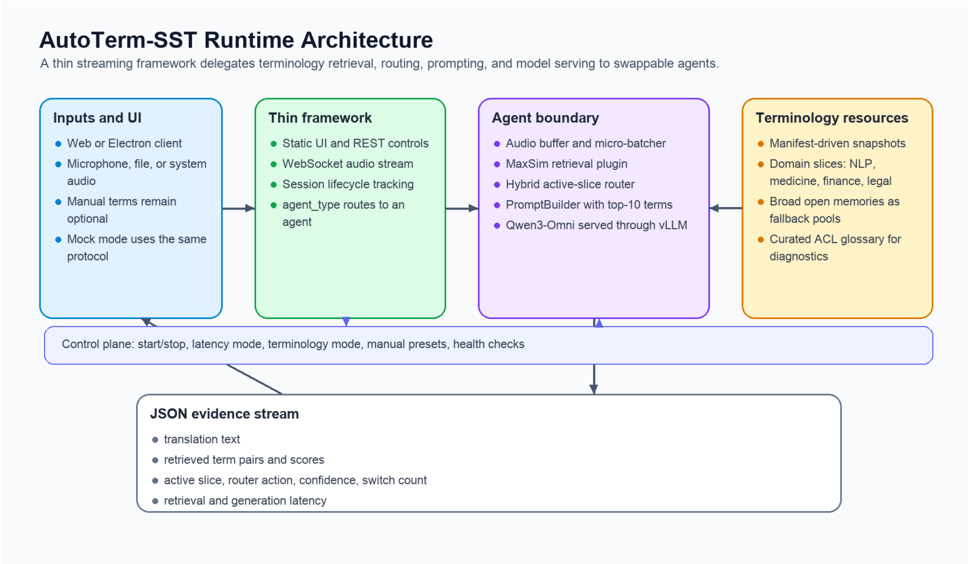

Figure 1: AutoTerm-SST separates transport/session logic from model-specific agent logic. Retrieval, active glossary routing, prompting, and vLLM generation all live inside the agent boundary, while the framework exposes a stable REST/WebSocket protocol and JSON evidence stream to the UI.

4 Adaptive Terminology Memory

AutoTerm-SST separates terminology inventory from the prompt interface. The *open memory* is a large offline resource derived from Wikidata/Wikipedia and curated glossaries, stored as manifest-driven snapshots with precomputed MaxSim indexes. The online system first retrieves at most 10 glossary candidates per chunk from the active inventory, then filters low-confidence matches before constructing the prompt. The automatic strategy controls which inventory slices are active and how candidates are ranked before filtering.

Open memory snapshots. Each snapshot maps a preset id to term files, metadata, and a MaxSim index. The runtime never stores large indexes in the code repository; instead, manifests point to a runtime root on the demo host. Current automatic presets are domain-specific slices such as NLP, medicine, finance, and legal; a curated 238-term ACL glossary is used as an oracle diagnostic, and broad open memories with 100k and 1M terms are used as rescue pools or scale diagnostics.

Hybrid active glossary router. The default live router is `hybrid_window_topic`. It is not a trained topic classifier and does not run a separate ASR path or source-transcript classifier in production. Instead, it combines signals that are already available in the streaming agent: generated target-translation topic windows when present, routing-only speech-window domain probes, weak speech-centroid evidence, and retrieved-reference metadata:

$$
\begin{aligned}
score(d) =& \lambda_t \, \mathrm{topic}_d(y_{\mathrm{recent}}) \\
&+ \lambda_p \, \mathrm{probe}_d(x_{\mathrm{speech}}) \\
&+ \lambda_c \, \cos(\mathrm{EMA}(q_{\mathrm{speech}}), c_d) \\
&+ \lambda_m \, \mathrm{vote}_d(\mathrm{references}),
\end{aligned} \tag{1}
$$

where y_{recent} is the recent generated translation window, x_{speech} is the recent speech window used for a routing-only domain probe, q_{speech} is the pooled speech query embedding, c_d is a domain centroid built offline from a slice's MaxSim text embeddings, and reference votes come from metadata such as source preset and domain. The default weights are $\lambda_t = 0.60$, $\lambda_p = 0.25$, $\lambda_c = 0.10$, and $\lambda_m = 0.05$; weights are renormalized over the signals available in a window, so routing already works before the first translations are generated. The older `embedding_refs` router remains available as a compatibility policy.

Conservative switching. The router is a policy for selecting active domain slices, not an open-world topic detector. It switches only after warmup, minimum update interval, confidence threshold, top-vs-runner-up margin, repeated consistent windows, domain-probe guards, and target index readiness. Generated target text requires multiple consecutive consistent windows; audio/probe-only switching is allowed only under stricter raw probe floors (Appendix A). Generic generated target text without positive topic evidence cannot dilute a strong probe or convert a weak probe into a high-confidence switch. Cold index loads are non-blocking: the agent preloads a target index in the background and activates it only after the retrieval plugin reports that the index is ready.

Capped prompt candidates. The automatic mode retrieves candidates from active slices, optionally adds `open_wiki_100k` under low-confidence fallback conditions, deduplicates by canonical source term, and caps retrieval at `PROMPT_K` $= 10$. The prompt may receive fewer than 10 candidates after score filtering; the system does not insert filler terms to reach the cap. Thus slice size trades coverage against ranking difficulty inside the top-10 retrieval cap: the 10k slice size reflects the clean bilingual terminology available per domain from our mining rather than a system limit, and broader inventories remain available as fallback pools and scale diagnostics (Appendix B). The live evidence metadata records active slices, candidate-pool size, prompt-reference counts, fallback events, and retrieval-quality metrics.

5 Demo Interface

The demo is designed for three audiences: researchers inspecting terminology errors in streaming SST, lecture or conference users who do not want to prepare glossaries, and developers testing new speech translation agents behind a stable protocol. The same interface can run in mock mode without a GPU or in live GPU mode with Qwen3-Omni and MaxSim retrieval.

Live controls. Users choose a model agent, language pair, latency multiplier, and terminology mode. In `auto_working` mode, the session starts from a configured domain-specific slice and may switch to another routed domain slice; every chunk keeps the same capped top-10 prompt interface. Manual presets and ad-hoc term overrides remain available as an advanced option; they inject into the prompt but never bias the router.

Evidence panel. For each partial translation, the UI renders the latest non-empty retrieved references as term-translation pairs with scores and source labels. A status line reports active memory backend, term count, and retrieval/generation latency; the adaptive row adds mode, inferred topic, confidence, active glossary, switch count, and router action. This makes terminology behavior visible rather than hidden inside the model prompt.

Packaging. The repository includes a mock-friendly launcher for UI/protocol development and a GPU launcher for the real RASST agent. The mock mode exercises the same REST/WebSocket flow and JSON evidence protocol, enabling reviewers to test the demo without model weights or GPUs. The live demo additionally uses a resident vLLM process, a dedicated retriever GPU, and an optional public tunnel.

Availability and licensing. The demo code is released under the MIT license on the GitHub repository, which is also the downloadable demo package: a source checkout in mock mode starts the same UI and protocol without GPUs. Live Qwen3-Omni mode requires A6000-class GPUs and the external RASST retriever checkpoint; model weights and Wikidata/Wikipedia-derived terminology resources retain their upstream licenses. A screencast video and a hosted live demo endpoint accompany this submission.[1]

6 Evaluation

Setup. We evaluate the terminology-memory behavior of the live AutoTerm-SST framework on A6000 GPUs. The RASST agent runs Qwen3-Omni with in-process vLLM tensor parallelism over two GPUs and a MaxSim retriever on a third GPU. Audio is from ACL6060 English-to-Chinese prepared segments. Unless otherwise stated, results use the framework JSON WebSocket protocol, which exposes translated text, retrieved references, active glossary metadata, and per-chunk retrieval latency. We separate completed system evidence into three probes: a terminology smoke comparison, a warm-retrieval scale sweep, and a mixed-domain routing probe.

Metrics. For terminology quality, we report term accuracy over an independent 9-term gold set for the smoke talk `2022.acl-long.268`. Masked BLEU removes target-side glossary translations from both hypothesis and reference; we use it only as a non-terminology sanity check, not as the primary glossary metric. For runtime, we report retrieved references per partial chunk and retrieval latency percentiles. For routing, we report switch

[1] Video and live demo: `https://github.com/luojiaxuan/autoterm-sst#demo`.

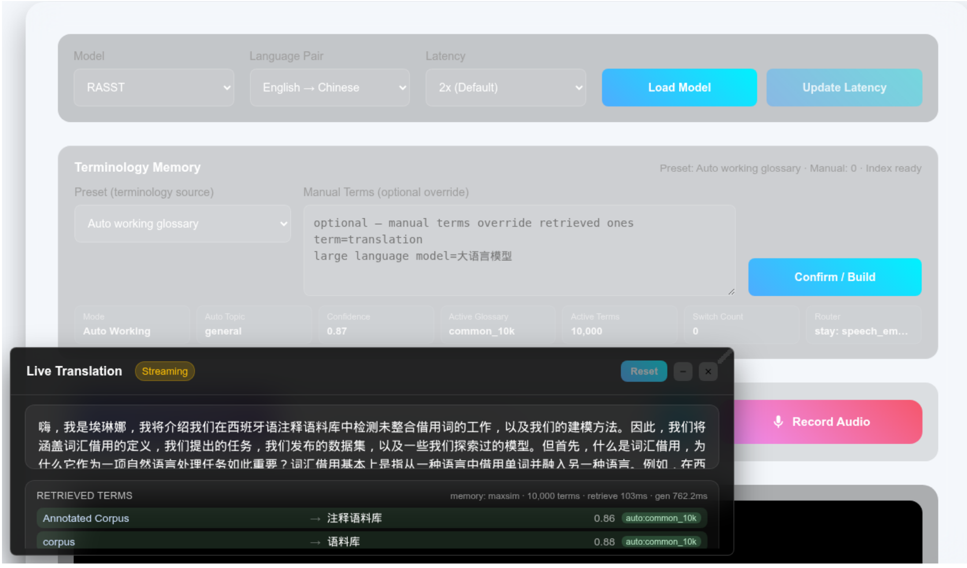

Figure 2: AutoTerm-SST web UI with live terminology evidence. The same interface controls model, latency, and terminology mode while rendering translated text, retrieved term evidence, active glossary status, router confidence, active term count, and retrieval/generation latency from the JSON WebSocket metadata.

Setting	Effort	Terms	Term acc.	M-BLEU	Refs	p95
None	0	0	0.444	42.34	0.00	–
Broad open	0	100k	0.444	45.29	5.03	111.54
Auto working	0	10k	**0.889**	45.80	2.29	88.20
Curated ACL	high	238	0.778	**47.07**	1.94	81.13

Table 1: Terminology smoke evaluation on the first eight audio segments of `2022.acl-long.268`, streamed as 31 JSON partial chunks. M-BLEU is masked BLEU; Refs is retrieved references per chunk; p95 is retrieval latency in ms. The zero-setup `auto_working` setting improves 9-term accuracy while injecting fewer references than the broad 100k memory.

Memory	Terms	p50/p95 ms	Refs/chunk
Curated	238	61.3 / 105.3	2.33
Domain	10k	78.5 / 82.1	1.80
Open	100k	64.1 / 82.6	4.89
Stress	1M	78 / 95	9.62

Table 2: Warm steady-state MaxSim retrieval latency across terminology memory sizes. Cold loading grows with index size (about 0.5s for 238 terms, 6.8s for 100k, and over 30s for 1M), so AutoTerm-SST preloads candidate working indexes outside the streaming batch path.

latency, wrong switches, and steady-state active-domain accuracy after transition grace windows. A separate 32-session stress run measures serving behavior under concurrent load; the glossary tables are terminology and retrieval evidence, not a replacement for token-delay metrics such as StreamLAAL (Ma et al., 2020).

Terminology usefulness. Table 1 compares four user-facing settings. Broad open memory is zero-effort, but it retrieves more low-precision candidates without improving term accuracy on this slice. The adaptive working glossary keeps the same zero user effort while selecting candidates from the domain-specific active inventory; it recovers eight of nine gold terms and reduces retrieval p95 relative to the broad 100k memory. The curated ACL glossary has the best masked BLEU and lowest retrieval latency, but it requires a manually curated domain glossary. Two caveats apply. First, the gold set is deliberately small, so we read Table 1 as smoke evidence; a five-talk streaming sweep against a 142-term technical gold set shows the same scale pattern, with term accuracy stable at 0.965–0.979 while retrieval precision falls from 0.48 (curated) to 0.14 (100k) as the inventory grows (Appendix B). Second, this short clip does not exercise a domain switch: the `auto_working` run stays on the configured initial slice because the router's confidence and interval guards do not fire; routing is evaluated separately below.

Scale and prompt noise. Table 2 shows that warm MaxSim retrieval stays near 60–105 ms (p50–p95) from curated glossaries to million-term memories. The bottleneck is not per-chunk exact retrieval; it is cold index loading and reference noise. As memories grow, refs/chunk increases, so the prompt receives more distractor terms even when useful terms remain retrievable. This supports the central design choice: keep the live active glossary

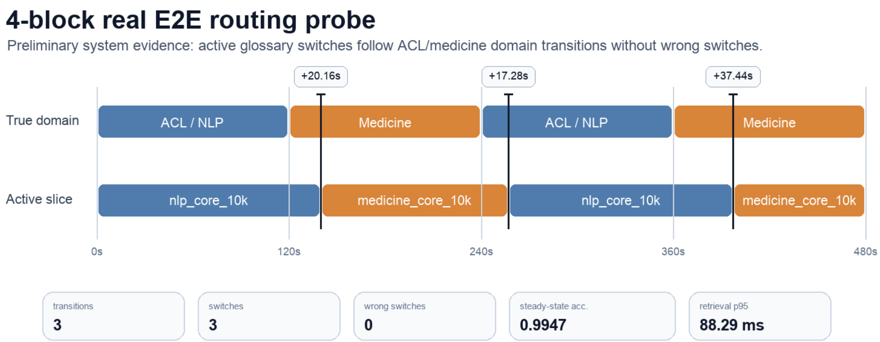

Figure 3: Mixed-domain routing probe on a 480s real E2E streaming playlist with ACL, medicine, ACL, and medicine blocks. The automatic active glossary switches after each true domain boundary with no wrong switches; the final switch is delayed because the medicine segment begins with speaker and session framing before explicit oncology terminology appears.

compact and treat large open memory as an offline resource pool and stress test.

Mixed-domain routing. Figure 3 reports the current 4-block real E2E routing probe for `auto_working`. The run observes three domain transitions, makes three active-slice switches, and records zero wrong switches. The first two switches occur 20.16s and 17.28s after the domain boundary; the final switch occurs 37.44s after the boundary because the opening of that medicine block contains general session framing before clear oncology terms. The run keeps the top-10 prompt invariant with 0.9947 steady-state active-domain accuracy (30s tolerance). A medicine-only session started from the default NLP slice recovers by switching to the medicine slice 59.5s into the session with zero wrong switches, so the initial slice is a warm start rather than a hidden oracle. The speech probe alone is noisy (0.52 top-1 accuracy), so generated-target evidence leads routing and the probe acts as a guard. A 640-window state-machine benchmark with controlled probe evidence passes all 16 alternating/random domain transitions with zero wrong switches; inverted- and contested-probe diagnostics fail as designed. Under benchmark-aligned union inventories, zero-setup routing also wins on terminology, exceeding *both* single-domain fixed inventories on a mixed playlist (0.916 vs. 0.839/0.825 combined term accuracy; Appendix C).

Concurrent streaming. In a 32-session stress test that streams live ACL audio to every session (`auto_working`, 300s each, real-time feed), all 32 sessions complete with zero failures or disconnects, median first-output latency rises only from 0.35s (single session) to 0.54s, and every session sustains the real-time pace of the input through vLLM continuous batching. This validates the transport and serving path under concurrent use while the terminology evaluation isolates active-memory behavior.

7 Limitations and Conclusion

AutoTerm-SST is a live terminology-aware SST system, not a new foundation model or topic classifier. The live demo serves En→Zh/Ja/De; routing stays correct across languages but switches more slowly without language-tuned topic keywords (Appendix D). Large open memories improve coverage but reduce retrieval precision under the 10-candidate cap, so the system routes among domain slices and keeps broad memory as a fallback pool. The hybrid router is intentionally conservative and may stay on the initial domain slice when evidence is ambiguous, as in our smoke talk. Our evaluations are single-run system probes; they characterize the live system rather than establish corpus-level significance. Finally, the real backend requires high-end NVIDIA GPUs and external model/retriever artifacts; mock mode supports UI and protocol testing without them.

We presented AutoTerm-SST, a zero-setup terminology-memory workbench for streaming speech translation. It keeps broad terminology resources offline, selects active inventory slices from streaming evidence, ranks candidates into a capped top-10 prompt interface, and exposes terms, glossary state, router decisions, and latency in real time — improving terminology recall with zero setup.

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A Router Configuration and Guards

Table 3 lists the default production configuration of the hybrid active-glossary router as released in `configs/autoterm_slices.yaml`; environment variables can override individual values per deployment. Guards are evaluated in order: warmup, update interval, target-equals-active, index readiness, probe floors, confidence, top-vs-runner-up margin, target-vs-current margin, post-switch cooldown, and consecutive-window consistency. Uncertain windows keep the current slice. The 640-window benchmark in Section 6 ran under the stricter three-window generated-target setting; the released default was later retuned to two windows with a 0.30 target-vs-current margin.

Parameter	Default
Signal weights (renormalized per window)	
generated-target topic λ_t	0.60
speech-window domain probe λ_p	0.25
speech centroid λ_c	0.10
reference metadata λ_m	0.05
Switch thresholds	
activation confidence	0.60
top-vs-runner-up margin	0.15
target-vs-current margin	0.30
consistent windows (text / gen. target / audio)	2 / 2 / 3
Audio-probe floors / schedule / retrieval	
probe top score / margin / positive domains	0.50 / 0.08 / 2
warmup / update / cooldown / staleness	20s / 10s / 30s / 120s
cap `PROMPT_K` / score threshold	10 / 0.78

Table 3: Default hybrid router configuration. Active inventory slices are typed: four domain slices (NLP, medicine, finance, legal; 10k terms each), a 100k open-memory rescue pool, a curated 238-term oracle (evaluation only), and a diagnostic common-terms slice.

B Glossary-Scale Streaming Sweep

Table 4 reports the five-talk streaming sweep referenced in Section 6: five concatenated ACL 60-60 talks (3,442s of audio) streamed through the live JSON WebSocket at latency multiplier 2, scored against a curated 142-term technical gold set. Each inventory always contains the full 238-entry curated ACL glossary; larger rows add Wikidata-mined AI/CS candidates as distractors. These are manual-preset runs isolating inventory scale (not `auto_working` runs). Term accuracy, BLEU, and masked BLEU stay in a narrow band up to 100k terms while retrieval precision collapses, motivating compact routed active slices.

C Mixed-Domain Terminology under Union Inventories

A glossary-channel comparison is only meaningful when the gold terms are inside the retrieval inventory: the broad wiki slices cover 29/142 of the ACL technical gold and 1/54 of the `medicine_606` gold, so their term accuracy mostly measures base-model retention. We therefore evaluate with *benchmark-aligned union inventories* at a fixed 10k denominator: the byte-identical gold glossary (238 ACL / 212 hard-medicine terms) plus seeded wiki fillers, mirroring the recipe of Appendix B (gold coverage 142/142 and 54/54).

Inventory	Terms	Term acc.	BLEU	M-BLEU	Gold@10	Prec@10	Refs
Curated ACL	238	0.972	58.27	53.10	0.972	0.480	1.53
+AI transl. 10k	10,238	0.965	58.72	53.35	0.972	0.234	3.09
+AI broad 10k	10,238	0.972	58.38	52.75	0.979	0.367	2.00
+AI broad 50k	50,238	0.979	58.65	53.24	0.979	0.196	3.64
+AI broad 100k	100,238	0.965	58.83	53.32	0.979	0.137	5.03

Table 4: Five-talk streaming sweep (latency multiplier 2) against the 142-term technical gold set. Gold@10 is the fraction of gold terms surfaced within the top-10 retrieval cap; Prec@10 is retrieval precision; Refs is surviving prompt references per chunk. At latency multiplier 1 the same pattern holds (term accuracy 0.944–0.958; precision 0.479 → 0.133).

Table 5 compares user-facing settings on a mixed ACL → medicine → ACL playlist (600s per block, live streaming at latency multiplier 2). Zero-setup `auto_working` exceeds *both* single-domain fixed inventories on the combined gold: it composes the right inventory per segment, switching twice with zero wrong switches (37.4s / 25.0s after the boundaries; steady-state active-domain accuracy 1.0; retrieval p95 101.6 ms).

Setting	$\mathrm{Acc}_{\mathrm{tech}}$	$\mathrm{Acc}_{\mathrm{raw}}$	ACL	Med.	Med. types	BLEU	M-BLEU
None	0.769	0.779	0.806	0.689	9/23	23.14	20.96
Fixed NLP	0.839	0.867	0.888	0.733	11/23	23.37	20.32
Fixed medicine	0.825	0.819	0.776	0.933	20/23	22.92	20.29
Auto (zero-setup)	**0.916**	**0.912**	**0.898**	**0.956**	**21/23**	**24.07**	20.49
Medicine oracle	0.811	0.801	0.776	0.889	18/23	22.94	20.59

Table 5: Mixed-domain streaming comparison with union inventories. $\mathrm{Acc}_{\mathrm{tech}}$ scores the curated technical gold (142 ACL + 23 medicine types), $\mathrm{Acc}_{\mathrm{raw}}$ the full raw annotation (238 ACL); per-domain columns use the technical view. "Medicine oracle" retrieves from the 212 gold terms only. The zero-setup router beats both fixed single-domain inventories on combined term accuracy without hurting BLEU.

D Multilingual Terminology (En → Ja/De)

The terminology resources are trilingual: the gold glossaries carry zh/ja/de translations, a single MaxSim index serves every target language (translations are resolved from the term list at prompt time), and the language pair selects the per-language demo model. Table 6 repeats the mixed full-length ACL → medicine → ACL comparison for En→Ja and En→De. Zero-setup routing has the best combined term accuracy in every language, under both gold denominators, and all switches across all three languages are directionally correct with zero wrong switches. Switch latency grows without language-tuned topic keywords (first/second switch: zh 37.4/25.0s; ja 115.3/125.6s; de 875.6/39.2s): the keyword channel is en/zh-tuned, so ja/de routing leans on the slower probe channel. Late switches cost some medicine coverage (de: 38/49 vs. 43/49 types) and some BLEU (ja), yet combined term accuracy still favors the router; language-tuned keyword lists are mechanical future work. German term matching is case-folded with light stem tolerance (exact matching undercounts under inflection); short variants keep strict word boundaries.

	Setting	Acc_{tech}	Acc_{raw}	ACL	Med.	Med. types	BLEU	M-BLEU
Ja	None	0.703	0.720	0.730	0.687	25/53	43.55	41.60
	Fixed NLP	0.718	0.760	0.900	0.608	21/53	44.19	42.21
	Fixed medicine	0.861	0.837	0.720	**0.946**	**48/53**	**44.90**	**42.83**
	Auto (zero-setup)	**0.902**	**0.894**	**0.930**	0.886	43/53	39.68	37.67
De	None	0.745	0.740	0.740	0.747	24/49	37.52	37.47
	Fixed NLP	0.762	0.768	0.840	0.720	23/49	37.21	36.91
	Fixed medicine	0.840	0.812	0.670	**0.934**	**43/49**	**37.63**	**37.18**
	Auto (zero-setup)	**0.897**	**0.869**	**0.880**	0.907	38/49	35.90	35.37

Table 6: En→Ja/De mixed-domain comparison (full-length playlist; ja: 266/350 technical/raw gold occurrences, 53 medicine types; de: 282/366, 49 types). The zero-setup router wins combined term accuracy in both languages; BLEU dips reflect the slower switches.